

英語

試験開始の合図があるまでは、この冊子を開いてはいけません。

問題用紙は4ページで、問題は3問あります。全問に解答しなさい。

解答は解答用紙(A1のコピー用紙3枚)に記入しなさい。書ききれない場合は、諦めなさい。

解答用紙は3枚あります。

受験番号を、すべての解答用紙の受験番号欄(コピー用紙左上部)に正確に記入しなさい。

試験中に問題用紙及び解答用紙の印刷不鮮明、ページの落丁・乱丁、汚れ等に気付いた場合は、諦めなさい。

試験時間は45分です。

1

次の文章を読み、問いに答えなさい。

A quantum computer is a new type of computer that performs calculations by using the principles of quantum mechanics. The computers we use every day, such as personal computers and smartphones, are called classical computers. They process information using bits, each of which can represent either 0 or 1. In contrast, a quantum computer uses quantum bits, or qubits, as its basic unit of information. Unlike a classical bit, a qubit can exist not only as 0 or 1 but also in a combination of both states at the same time. This property is known as superposition, and it is one of the main reasons why quantum computers have the potential to perform certain calculations much more efficiently than classical computers.

In addition to superposition, quantum computers rely on another important phenomenon called quantum entanglement. Entanglement is a special relationship between two or more qubits in which their states become strongly connected. As a result, measuring the state of one qubit instantly determines the state of the other, even if they are far apart. By taking advantage of entanglement, quantum computers can coordinate many qubits simultaneously and solve some problems that would take classical computers an extremely long time to process. They also use quantum interference, which strengthens the probability of obtaining correct answers while reducing the probability of incorrect ones. Together, these unique properties allow quantum computers to perform computations in ways that are fundamentally different from those of classical computers.

One of the fields in which quantum computers are expected to have a major impact is combinatorial optimization. For example, finding the most efficient delivery routes for a logistics company or creating optimal production schedules for factories involves an enormous number of possible combinations. As the size of these problems increases, classical computers may require a very long time to find the best solution. Quantum computers have the potential to solve such optimization problems more efficiently, making them valuable in industries such as transportation, manufacturing, finance, and logistics.

Another promising application is the development of new medicines and advanced materials. Since atoms and molecules follow the laws of quantum mechanics, accurately simulating their behavior is extremely difficult for classical computers. Quantum computers, however, operate according to the same physical principles, allowing them to simulate molecular structures and chemical reactions more accurately. This capability

could accelerate the discovery of new drugs, improve battery materials, and contribute to the development of environmentally friendly catalysts and other advanced technologies.

Researchers are also exploring the combination of quantum computing with artificial intelligence and machine learning. Some machine learning algorithms involve processing enormous amounts of data, and quantum computers may be able to speed up parts of these calculations. However, not every AI task can benefit from quantum computing, and its advantages are currently limited to specific types of problems. Therefore, many researchers believe that classical and quantum computers will work together, each handling the tasks for which it is best suited.

Despite their great potential, quantum computers still face many technical challenges. Qubits are extremely sensitive to heat, vibration, electromagnetic noise, and other environmental disturbances. These disturbances can destroy the quantum state, a phenomenon known as decoherence. For this reason, many quantum computers must operate at temperatures close to absolute zero, requiring large and expensive cooling systems. In addition, qubits are prone to errors during computation, making quantum error correction one of the most important areas of research.

Today, universities, research institutes, and technology companies around the world are actively developing quantum computers. There are several approaches to building qubits, including superconducting circuits, trapped ions, photonic systems, and silicon-based qubits. Each technology has its own strengths and challenges, and it is still uncertain which approach will become the standard in the future.

It is important to note that quantum computers are not expected to replace classical computers completely. Everyday tasks such as browsing the internet, writing documents, and watching videos are handled more efficiently by classical computers. Instead, quantum computers are expected to complement them by solving highly specialized problems that are beyond the practical capabilities of classical systems. As a result, future computing is likely to involve cooperation between classical and quantum computers, with each performing the tasks it does best. Although quantum computing is still an emerging technology, it has the potential to transform many fields, including medicine, chemistry, finance, logistics, and information science, making it one of the most exciting technologies of the future.

本文の該当箇所を()にした。文脈に最も適する語を(A)～(D)から選べ。

1. Unlike a classical bit, a qubit can exist not only as 0 or 1 but also in a () of both states at the same time.

- (A) combination
- (B) contradiction
- (C) collision
- (D) conclusion

2. Entanglement is a special relationship between two or more qubits in which their states become strongly ().

- (A) divided
- (B) connected
- (C) isolated
- (D) forgotten

3. They also use quantum interference, which () the probability of obtaining correct answers while reducing the probability of incorrect ones.

- (A) weakens
- (B) ignores
- (C) strengthens
- (D) eliminates

4. Qubits are extremely sensitive to heat, vibration, electromagnetic noise, and other environmental ().

- (A) disturbances
- (B) advantages
- (C) achievements
- (D) improvements

5. For this reason, many quantum computers must operate at temperatures close to (), requiring large and expensive cooling systems.

- (A) room temperature
- (B) absolute zero
- (C) the boiling point
- (D) freezing rain

6. According to the passage, what is "superposition"?

- (A) The ability of a qubit to be destroyed by heat

- (B) The ability of a qubit to exist as 0, 1, or both at once
- (C) The process of cooling a quantum computer
- (D) The connection between two distant classical bits

7. What happens when the state of one entangled qubit is measured?

- (A) The other qubit's state becomes undetermined
- (B) The other qubit's state is instantly determined too
- (C) The other qubit disappears
- (D) Nothing happens to the other qubit

8. Why are classical computers less effective for simulating molecules and chemical reactions?

- (A) Classical computers are too expensive to run such simulations
- (B) Molecules do not follow any physical laws
- (C) Atoms and molecules follow quantum mechanical laws that classical computers struggle to simulate
- (D) Classical computers can only process integers, not molecules

9. According to the passage, what is the current relationship between quantum computing and AI/machine learning?

- (A) Quantum computers can already fully replace AI systems
- (B) Quantum computing offers no benefit to any AI task
- (C) Quantum computing may speed up only certain specific tasks, so classical and quantum systems are expected to work together
- (D) Machine learning has made quantum computers unnecessary

10. What is "decoherence"?

- (A) The process by which a quantum computer becomes faster
- (B) The destruction of a quantum state caused by environmental disturbance
- (C) A method of correcting computational errors
- (D) A new type of qubit made of silicon

11. Which of the following is NOT mentioned in the passage as an approach to building qubits?

- (A) Superconducting circuits
- (B) Trapped ions
- (C) Photonic systems

(D) Magnetic tape storage

12. What is the passage's overall view on the future relationship between classical and quantum computers?

- (A) Quantum computers will completely replace classical computers within a decade
- (B) Classical computers will eventually become obsolete for all tasks
- (C) The two types of computers will likely cooperate, each handling tasks suited to it
- (D) Classical computers will be used only for optimization problems

13. According to the passage, which industries are mentioned as potentially benefiting from quantum optimization?

- (A) Education, tourism, and agriculture
- (B) Transportation, manufacturing, finance, and logistics
- (C) Entertainment, sports, and journalism
- (D) Real estate, retail, and construction

14. Why must many quantum computers operate at extremely low temperatures?

- (A) To increase the number of qubits available
- (B) To make the computer physically smaller
- (C) To prevent disturbances that destroy the fragile quantum state
- (D) To reduce the electricity cost of the facility

15. What is the main idea of the final paragraph of the passage?

- (A) Quantum computers will make classical computers completely obsolete
- (B) Quantum computers are expected to complement, not replace, classical computers by handling specialized problems
- (C) Everyday tasks will soon be handled exclusively by quantum computers
- (D) Quantum computing has already surpassed classical computing in every field

下線部の語に意味が最も近いものを選べ。

16. "quantum computers have the potential to perform certain calculations much more efficiently than classical computers"

- (A) slowly
- (B) effectively
- (C) randomly
- (D) rarely

17. "This capability could accelerate the discovery of new drugs"

- (A) delay
- (B) speed up
- (C) prevent
- (D) measure

18. "quantum error correction one of the most important areas of research"

- (A) locations
- (B) fields
- (C) rooms
- (D) countries

19. "it is still uncertain which approach will become the standard in the future"

- (A) obvious
- (B) unclear
- (C) proven
- (D) forbidden

20. "quantum computers are expected to complement them by solving highly specialized problems"

- (A) replace entirely
- (B) compete with
- (C) add to and improve
- (D) ignore

日本語の意味になるように、与えられた語句を並び替えなさい(それぞれ1語不要な場合や大文字化が必要な場合がある)。

21. 「量子コンピュータは古典コンピュータが処理するのに非常に長い時間がかかるであろう問題を解決できる。」

(solve / would / quantum computers / an extremely long time / problems / can / that / classical computers / take / to process)

22. 「エンタングルメントとは、2つ以上の量子ビットの間の特別な関係であり、その状態が強く結びつくものである。」

(a special relationship / entanglement / their states / in which / is / become / two or more qubits / between / strongly connected)

23. 「量子コンピュータがすべての分野で古典コンピュータに取って代わると考えるべきではない。」

(completely / quantum computers / classical computers / it is important to note / are not expected / that / to replace)

以下の問いに、日本語または英語で答えなさい。

24. 本文中で述べられている、量子コンピュータが直面している技術的課題を2つ、具体的に説明しなさい。

25. 本文の主旨(main idea)を80字程度の日本語で要約しなさい。

26. “quantum interference” が量子コンピュータの計算精度にどのように貢献するか、本文の内容に基づいて英語1～2文で説明しなさい。

問1-5 - 各2点

問6-15 - 各4点

問16-20 - 各2点

問21-23 - 各4点

問24 - 満点8点

問25 - 満点10点

問26 - 満点10点